



BRB CAPITAL GROUP · RESEARCH & TECHNOLOGY

# Transparent Models, Constrained Machines

A Provenance-Bounded Architecture for AI-Assisted Equity Research on the Nigerian Exchange

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*The methodological thesis underlying the BRB NGX Analyst Platform*

Working Paper · v1.0 · August 2026 · Internal analytical tool — not investment advice

**Abstract.** We describe the design thesis of an internal decision-support platform for equity research on the Nigerian Exchange (NGX) operated by a dual-regulated (FCA UK / SEC Nigeria) fund and portfolio manager. The central argument is that trustworthy machine assistance in a frontier, information-scarce market is best achieved not by maximising model autonomy but by *bounding* it: pairing transparent, auditable statistical models with large-language-model (LLM) reasoning that is strictly constrained by data provenance and by compliance-by-construction. We set out six components: (i) a delayed-quote data model with market-hours-aware caching and exact reconstruction of plan-gated market aggregates; (ii) a reproducible Monte-Carlo geometric Brownian-motion (GBM) ensemble for probabilistic price scenarios with walk-forward validation; (iii) fundamental and dividend scenario models; (iv) a full single-benchmark portfolio-risk stack (beta, Jensen alpha, tracking error, information ratio, Sharpe, Sortino, historical VaR/CVaR, drawdown); (v) a provenance-constrained LLM layer in which the model never originates market figures and all position arithmetic is recomputed server-side; and (vi) a maker-checker governance workflow. We argue this architecture converts an opaque, hallucination-prone tool into an auditable analyst instrument, and we are explicit about its limitations.

**Keywords:** frontier markets · NGX · Monte-Carlo simulation · geometric Brownian motion · portfolio risk · large language models · data provenance · RegTech · decision support

## 1. Introduction and thesis

The Nigerian Exchange is a frontier equity market characterised by concentrated liquidity in a small number of large-capitalisation names, uneven disclosure, and market data that is *delayed* rather than streamed in real time. Analysts must therefore synthesise sparse, latency-bound information into portfolio decisions under a strict dual-regulatory regime. Two temptations arise when applying modern machine intelligence to this setting. The first is to let a statistical forecast masquerade as a prediction; the second is to let a language model both *reason about* and *supply* the underlying facts — inviting confident fabrication of prices, ratios and events.

Our thesis is that reliability in this environment is a property of **constraint**, not autonomy. Concretely, we hold every quantitative output to three standards — it must be transparent (reconstructable from inputs), reproducible (seeded and deterministic), and provenance-bounded (no figure originates inside a model that cannot cite it). The platform is consequently positioned as *decision support for a qualified professional*, never as automated advice, and this positioning is enforced in code and workflow rather than merely stated in a disclaimer. The remainder of this paper documents the models and the strategy that operationalise that thesis.

## 2. Design principles

- **Transparency.** Analytical logic lives in pure, framework-free, unit-tested functions; every displayed number is traceable to inputs and a documented formula. No result is a black box.
- **Provenance.** Market facts enter the system only through the authenticated data proxy or through documents the system has read and can cite. Reasoning components consume these facts; they never manufacture them.
- **Compliance-by-construction.** Illustrative-scenario labelling, delayed-quote disclosure with explicit freshness, a maker-checker approval gate, role-based access and a full audit trail are structural, not optional.
- **Reproducibility.** Stochastic components use a seeded pseudo-random generator, so a given input reproduces a given scenario fan exactly — a prerequisite for review and for regulatory defensibility.

## 3. Data model and market microstructure

The platform is, by design, a *delayed-quote terminal*. Prices republish on the order of every twenty minutes during exchange hours (Mon–Fri, 09:00–16:00 West Africa Time) and reflect the last close outside them; the interface surfaces an explicit ‘as of’ timestamp on all prices. A market-hours-aware cache assigns short time-to-live to live-price groups while the market is open (to capture each republish) and long time-to-live when it is closed (to conserve a shared monthly data quota), with static reference data cached for many hours.

Certain market aggregates (sector performance, breadth, year-to-date leaders) are gated behind higher data-plan tiers. Rather than estimate them, the platform *reconstructs* them exactly from the underlying constituent list on an equal-weighted basis and labels them as derived — a real calculation, not an approximation. Foreign-exchange handling is treated carefully because the vendor returns spot and historical rates under different inversion conventions; each call site normalises to naira-per-unit before display.

## 4. Probabilistic price forecasting: a Monte-Carlo GBM ensemble

Forward price scenarios are generated by an ensemble of geometric Brownian-motion paths rather than a single point forecast, so the output is a *distribution* of outcomes expressed as probability bands. Let  $S_t$  denote the closing price and  $r_t = \ln(S_t / S_{t-1})$  the daily log-return.

### 4.1 Drift and volatility estimation

The per-step drift  $d$  blends two views of trend and is damped to temper long-horizon extrapolation. Let  $d_{hist}$  be the mean historical log-return and  $d_{reg}$  the slope of an ordinary-least-squares fit of  $\ln(\text{price})$  on time over the most recent 504 observations. With damping factor  $k = 0.5$ ,

$$d = k \times (0.5 d_{hist} + 0.5 d_{reg}), \quad s = \text{stdev}(r_t)$$

where  $s$  is the daily volatility (standard deviation of log-returns). The blend gives weight to both the realised mean return and the fitted trend, while the damping prevents a steep recent slope from producing implausible multi-month projections.

### 4.2 Simulation

Each of  $P = 2000$  paths is evolved over  $H = 252$  trading days. Because  $d$  is already a mean *log*-return, each step draws the log-return directly as  $\text{Normal}(d, s)$  — no additional Ito correction is applied (applying one would bias the median low by one-half the daily variance per step):

$$S_t = S_{t-1} \times \exp(d + s \times Z_t), \quad Z_t \sim \text{Normal}(0, 1)$$

At each step the empirical cross-sectional percentiles of the 2000 simulated prices are recorded. The 10th, 25th, 50th, 75th and 90th percentiles form the fan chart; the median (P50) is the base case and the P10/P90 pair the bear/bull envelope, summarised at the 3-, 6- and 12-month horizons.

### 4.3 Validation and reproducibility

Model quality is reported alongside every forecast, not hidden. A walk-forward diagnostic re-fits the model on all but the last twenty-one trading days, projects the median forward, and reports the mean absolute percentage error (MAPE) against the realised price; the trend-fit R-squared is shown as a goodness-of-fit signal. Users are told plainly that higher backtest error and lower R-squared imply wider real-world uncertainty than the bands alone convey. All paths derive from a seeded generator, so the scenario fan is exactly reproducible.

**Assumptions and their limits.** GBM presumes independent, identically distributed, log-normal shocks and a stationary drift/volatility regime. Frontier-market returns exhibit fat tails, autocorrelation and regime shifts; thin liquidity can make realised paths jump. The forecast is therefore framed throughout as an illustrative scenario, not a prediction, recommendation or guarantee.

## 5. Fundamental and dividend scenario models

Alongside the statistical view, a fundamentals view derives an implied price from an earnings assumption. With trailing-twelve-month earnings per share (EPS) and an assumed exit price/earnings multiple  $m$ ,

$$\text{implied price} = \text{EPS}_{ttm} \times m$$

A sensitivity grid spans a range of growth and exit-multiple assumptions so the analyst can read the valuation surface rather than a single figure. The dividend model projects a forward path from the trailing-twelve-month dividend per share, grown at the historical compound annual growth rate clamped to a sane band (to prevent a single special dividend from exploding the projection), and reports a yield-on-cost path. All inputs are sourced from the market data feed or from cited filings; none are supplied by a model.

## 6. Portfolio construction and analytics

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Portfolios are specified in either target-weight or share-unit mode; a common effective-weight function values unit-mode books by units multiplied by price so that both representations feed one analytics path. On construction the platform computes a weighted dividend yield, a weighted price/earnings ratio (only positive earners contribute, since a negative EPS would corrupt the aggregate) and a sector allocation. An illustrative backtest rebases the blended book and its benchmark to 100 and holds current weights constant over the window — without rebalancing, dividends or transaction costs — reporting cumulative return, annualised volatility and maximum drawdown. The constant-weight, cost-free nature of this backtest is disclosed as a deliberate simplification.

## 7. Portfolio risk analytics

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Risk statistics are computed from aligned daily *level* series (the rebased portfolio and benchmark), from which returns are derived internally. Writing  $p$  and  $b$  for portfolio and benchmark daily returns and  $r_f$  for the daily risk-free rate, the platform reports systematic and risk-adjusted measures:

$$\text{beta} = \text{cov}(p, b) / \text{var}(b), \quad \text{alpha} = [ \text{mean}(p) - (r_f + \text{beta} ( \text{mean}(b) - r_f ) ) ] \times 252$$

$$\text{tracking error} = \text{stdev}(p - b) \times \text{sqrt}(252), \quad \text{Sharpe} = \text{annualised excess} / \text{annualised volatility}$$

Jensen's alpha is the annualised return in excess of the capital-asset-pricing-model expectation; the information ratio scales active return by tracking error; the Sortino ratio replaces total volatility with downside deviation measured against the risk-free target. Tail risk is estimated non-parametrically: the one-day 95% historical Value-at-Risk is the empirical loss quantile, and the 95% conditional VaR (expected shortfall) is the mean loss beyond it — a coherent tail measure. A holding-level correlation matrix exposes diversification and concentration. Maximum drawdown is measured peak-to-trough on the level series.

## 8. Positions accounting and technical indicators

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Booked transactions drive an average-cost engine that tracks running units, cost basis and realised profit and loss, with fees pro-rated to matched units on partial sells. Technical overlays — simple moving averages, the 14-period relative-strength index, drawdown-from-peak and annualised volatility — are computed on the full price history and then sliced to the visible window, so that long-lookback indicators remain correct on short views.

## 9. The reasoning layer and the provenance strategy

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The distinctive element of the architecture is how it uses large language models. The governing rule is simple and absolute: **the model is a reasoner over supplied evidence, never a source of market facts**. NGX prices, fundamentals and ratios are injected from the authenticated data feed or extracted from official filings that the system has downloaded and read; the model is instructed to use only the figures provided, and its outputs are returned as validated structured objects rather than free text where a schema applies.

Two further safeguards close the loop. First, **arithmetic is never trusted to the model**: all position sizing and profit-and-loss are recomputed server-side from units and live prices, any instrument the model invents is dropped, and entry prices are reconciled to live quotes. Second, filing extractions carry **page-level citations** back to the source document, so every extracted figure is one click from its origin. Together these convert the model from a potential fabricator into an auditable analyst that must show its work.

### 9.1 Reasoning-layer features

- **Daily desk briefing** — a structured read of the session (headline, market action, movers and sectors, FX, bottom line) written strictly from the day's own market data, with optional cited web context for qualitative events.
- **Filing extraction** — income statement, balance sheet and cash-flow tables lifted from official NGX filing PDFs with page citations.
- **Portfolio review** — a cash-constrained CIO analysis (diagnosis, trims, adds sized in naira, reconciled reallocation) whose resulting trades are routed to the approval workflow rather than executed.
- **AI portfolio builder** — a liquidity-first CIO/head-trader construction for a given budget and horizon, with defined entry, target and stop levels, sized server-side to live prices and saveable as a tracked paper portfolio.
- **Signal, dividend and sector-momentum reads** — structured buy/hold/sell theses and outlooks framed as analysis, not instruction.

The reasoning layer is model-plural: a high-capability model serves the analytical narratives and a specialised reasoning model serves the portfolio-construction task, each with refusal handling and graceful degradation to a 'not configured' state when credentials are absent.

## 10. Validation, governance and compliance

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Quantitative correctness is defended by a suite of unit tests over the pure analytical functions, and forecast quality by the walk-forward error reported in-product. Governance is structural. The platform is operated under dual FCA (UK) and SEC (Nigeria) oversight, and its posture is enforced in code: every forecast carries an illustrative-scenario label; performance displays carry the standard past-performance caveat; rebalancing output is an analytical suggestion that must pass a **maker-checker** approval — an analyst submits, and a different principal fund manager approves, with self-approval blocked. Access is role-based, secrets are encrypted at rest, and portfolio, transaction and approval actions are written to an immutable audit trail. Crucially, the human remains in the loop by construction: outputs are decision support for the judgement of a qualified professional, and no output constitutes investment advice or an order.

## 11. Limitations and future work

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- The GBM forecast assumes normal, i.i.d. shocks and a stationary regime; jump/regime-switching dynamics and fat tails are not modelled. Future work: regime-switching and jump-diffusion variants.
- Risk is single-benchmark (CAPM-style beta); a multi-factor risk model would better attribute frontier-market risk.
- Data are delayed, not streamed; the platform is explicitly a delayed-quote terminal. A real-time feed would require a different vendor tier.
- The illustrative backtest omits rebalancing, dividends and transaction costs by design.
- LLM reasoning, though provenance-bounded, still requires human verification of the qualitative narrative.

■ The embedded single-instance datastore is appropriate for a single desk; horizontal scale would migrate to managed Postgres behind the same standard-SQL repository layer.

## 12. Conclusion

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The models in this platform are individually conventional — geometric Brownian motion, CAPM-style risk, average-cost accounting, discounted fundamentals. The thesis is not in any single model but in the discipline that surrounds them: transparency so that every number is reconstructable; reproducibility so that every scenario is repeatable; and a provenance boundary so that machine reasoning augments, but never fabricates, the evidence base. In a frontier market with delayed, sparse data and a demanding compliance regime, we argue that this bounded, auditable design — rather than maximal model autonomy — is what makes AI-assisted equity research trustworthy enough to sit on an analyst’s desk.

## Selected methodological references

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This is an internal research and methodology white paper describing the design of the BRB NGX Analyst Platform. It is not investment advice, a research recommendation on any security, or a solicitation. Forecasts are illustrative scenarios; prices are delayed; past performance does not indicate future results.